G.L.A UNIVERSITY , MATHURA,U.P.



GLA UNIVERSITY

PRACTICAL FILE

SUBMITTED TO: MR. ASHUTOSH SIR

SUBMITTED BY: GOVIND SHARMA

SECTION : G

UNIVERSITY ROLL NO.:2115000419

CLASS ROLL NO. : G-24

BRANCH : BTECH CSE

SUBJECT : DBMS LAB

FACULTY NAME : COMPUTER SCIENCE & ENGINEERING

GOVIND SHARMA

EXPERIMENT – 1

```
create table College(
cName varchar2(10),
state varchar2(10),
enrollment int);

create table Student(
sID int,
 sName varchar2(10),
 GPA number(2,1),
 sizeHS int,
 DoB date);

create table Apply(
sID int,
 cName varchar2(10),
 major varchar2(20),
 decision char(1));

insert into Student values (123, 'Amy', 3.9, 1000, '26-JUN-96');
insert into Student values (234, 'Bob', 3.6, 1500, '7-Apr-95');
insert into Student values (345, 'Craig', 3.5, 500, '4-Feb-95');
insert into Student values (456, 'Doris', 3.9, 1000, '24-Jul-97');
insert into Student values (567, 'Edward', 2.9, 2000, '21-Dec-96');
insert into Student values (678, 'Fay', 3.8, 200, '27-Aug-96');
insert into Student values (789, 'Gary', 3.4, 800, '8-Oct-96');
insert into Student values (987, 'Helen', 3.7, 800, '27-Mar-97');
insert into Student values (876, 'Irene', 3.9, 400, '7-Mar-96');
insert into Student values (765, 'Jay', 2.9, 1500, '8-Aug-98');
insert into Student values (654, 'Amy', 3.9, 1000, '26-May-96');
insert into Student values (543, 'Craig', 3.4, 2000, '27-Aug-05');

insert into College values ('Stanford', 'CA', 15000);
insert into College values ('Berkeley', 'CA', 36000);
insert into College values ('MIT', 'MA', 10000);
insert into College values ('Cornell', 'NY', 21000);
insert into College values ('Harvard', 'MA', 50040);

insert into Apply values (123, 'Stanford', 'CS', 'Y');
insert into Apply values (123, 'Stanford', 'EE', 'N');
insert into Apply values (123, 'Berkeley', 'CS', 'Y');
insert into Apply values (123, 'Cornell', 'EE', 'Y');
insert into Apply values (234, 'Berkeley', 'biology', 'N');
insert into Apply values (345, 'MIT', 'bioengineering', 'Y');
insert into Apply values (345, 'Cornell', 'bioengineering', 'N');
insert into Apply values (345, 'Cornell', 'CS', 'Y');
insert into Apply values (345, 'Cornell', 'EE', 'N');
```

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```
insert into Apply values (678, 'Stanford', 'history', 'Y');
insert into Apply values (987, 'Stanford', 'CS', 'Y');
insert into Apply values (987, 'Berkeley', 'CS', 'Y');
insert into Apply values (876, 'Stanford', 'CS', 'N');
insert into Apply values (876, 'MIT', 'biology', 'Y');
insert into Apply values (876, 'MIT', 'marine biology', 'N');
insert into Apply values (765, 'Stanford', 'history', 'Y');
insert into Apply values (765, 'Cornell', 'history', 'N');
insert into Apply values (765, 'Cornell', 'psychology', 'Y');
insert into Apply values (543, 'MIT', 'CS', 'N');
```

- select sname,dob from student;
- select sname from student where gpa>3.7;
- select sname from student where sizehs>=1000 and dob>'31-DEC-1996';
- select sname from student where gpa between 2.9 and 3.9;
- select * from college where state='MA';
- select sname from student where gpa>2.0 and gpa<3.5;
- select sname from student where dob>'1-jul-96' order by dob;
- select sid,cname,decision from apply where decision='Y';
- select sid,cname from apply where cname='Stanford';
- select cname from college where enrollment>1001;

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EXPERIMENT – 2

```
Create table department (
 deptno number(3),
 dname varchar2(20) unique,
 location varchar2(20) not null check( location in ('Agra', 'Delhi', 'Pune')));

Create table employee (
empno varchar2(5) primary key check (empno like 'E%'),
 ename varchar2(20) unique,
 Designationvarchar2(20) not null,
 salary number(10) check (salary between 15000 and 50000),
 dob date not null,
 dno number(3) references department);

Create table candidate (
Candidate_ID number(6) primary key,
 Candidate_Name varchar2(20) not null,
Candidate_Email varchar2(30) unique check(Candidate_Email like '%_@%_.%_'),
 Candidate_Dept varchar2(2) default 'HR',
 Manager_ID number(6) references candidate);
```

- alter table college add primary key(cname);
- alter table student Add primary key(sid);
- alter table apply Add primary key(sid,cname,major);
- alter table apply Constraint apply_fk foreign key(sid) references student;
- alter table apply modify(major varchar2(25));
- alter table apply add(decision varchar2(3) not null);
- alter table apply modify(decision char(1));
- alter table apply drop constraint apply_fk;
- alter table student drop column sizehs;
- alter table college drop primary key;

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EXPERIMENT – 3

- SELECT *FROM Student INNER JOIN Apply ON Student.sID=Apply.sID;
- SELECT Student.sID,Student.sName,Student.GPA,Apply.cName,Apply.major FROM Student INNER JOIN Apply ON Student.sID=Apply.sID;
- SELECT *FROM Apply INNER JOIN College ON Apply.cName=College.collegeName AND college.state='CA';
- SELECT Student.sID,Student.sName,Student.GPA,Apply.sID,Apply.cName FROM Student INNER JOIN Apply ON Student.sID=Apply.sID AND Student.GPA>3.7 AND Apply.cName='Stanford';
- SELECT *FROM Student INNER JOIN Apply ON Student.sID=Apply.sID AND Apply.major='CS' AND Apply.decision='N';
- SELECT *FROM Student INNER JOIN Apply ON Student.sID=Apply.sID INNER JOIN College ON Apply.cName=College.collegeName and college.state='NY';
- SELECT Student.sID, Student.sName, Student.GPA, Student.sizeHS FROM Student LEFT JOIN Apply ON Student.sID=Apply.sID WHERE Apply.sID IS NULL ;
- SELECT college.collegeName FROM College LEFT JOIN Apply ON College.collegeName=Apply.cName where Apply.cName IS NULL;
- SELECT sID FROM Apply GROUP BY sID HAVING COUNT(sID)=1;
- SELECT Student.sName, Student.GPA FROM Student INNER JOIN Apply ON Student.sID=Apply.sID INNER JOIN College ON Apply.cName=College.collegeName AND College.enrollment < 25000;

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EXPERIMENT – 4

- select count(*) from Student;
- select avg(GPA) from Student;
- select min(GPA) as "min_GPA" , max(GPA) as "max_GPA" from Student;
- select count(GPA) from Student where GPA>=3.7;
- select max(GPA),min(GPA),avg(GPA,sum(GPA) from Student;
- select count(cName) from Apply;
- select count(distinct(major)) from Apply;
- select count(*) from Apply;
- select avg(distinct(GPA)) from Student;
- select count(*) from Apply where decision='Y';

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EXPERIMENT – 5

- SELECT SID,SNAME FROM STUDENT WHERE SID IN (SELECT SID FROM APPLY WHERE MAJOR = ‘CS’);

- SELECT SID , SNAME FROM STUDENT WHERE SIZEHS IN (SELECT SIZEHS FROM STUDENT WHERE SNAME = ‘JAY’);

- SELECT SID , SNAME , FROM STUDENT WHERE SIZEHS IN (SELECT SIZEHS FROM STUDENT WHERE SNAME = ‘JAY’) AND SNAME NOT IN (‘JAY’);

- SELECT SID , SNAME , GPA FROM STUDENT WHERE GPA NOT IN (SELECT GPA FROM STUDENT WHERE SNAME = ‘IRENE’);

- SELECT CNAME FROM APPLY WHERE SID IN (SELECT SID FROM STUDENT SNAME LIKE ‘J%’);

- SELECT DISTINCT MAJOR FROM APPLY WHERE SID IN (SELECT SID FROM STUDENT WHERE SNAME = ‘IRENE’);

- SELECT DISTINCT SID,MAJOR FROM APPLY WHERE MAJOR IN (SELECT MAJOR FROM APPLY WHERE SID IN (SELECT SID FROM STUDENT S1 WHERE SNAME=’IRENE’));

- SELECT DISTINCT SID , MAJOR FROM APPLY WHERE MAJOR (SELECT MAJOR FROM APPLY WHERE SID IN (SELECT SID FROM STUDENT WHERE SNAME = ‘IRENE’)) AND SID NOT IN (SELECT SID FROM STUDENT WHERE SNAME = ‘IRENE’);

- SELECT COUNT (DISTINCT CNAME) FROM APPLY SID IN (SELECT SID FROM STUDENT WHERE SNAME = ‘JAY’);

- SELECT SID FROM APPLY WHERE CNAME IN (SELECT CNAME FROM APPLY WHERE SID IN (SELECT SID FROM STUDENT WHERE SNAME = ‘JAY’) ;

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EXPERIMENT – 6

- Alter table student add DoB date;
- Update student SET DOB = '7-APR-95' WHERE SID = 234;
- Select round(avg(gpa),2) from student;
- Select extract(year from dob) from Student where sizehs < 1000;
- Select extract(year from sysdate()) - extract(year from dob) from student;
- Select upper(sname),lower(cname) from student s,apply a where s.sid = a.sid;
- Select substr(sname,4,1) from student;
- Select sid,sname from student where length(sname) > 3;
- Select FLOOR(avg(gpa)),ceil(avg(gpa)),TRUNCATE(avg(gpa),1) from student;
- Select * from student where mod(sid,2) = 0;

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EXPERIMENT – 7

- Create or replace view WeakStudent as select sID, sName, GPA, sizeHS, DoB from student where GPA > 3.7;

- Create or replace view cView(collegeName, seats, state) as select cName, enrollment, state from student where GPA > 3.7;

- Create or replace view CSaccept as select sID,cName from apply where major = 'CS' and decision = 'Y';

- Create view CSberkeley as select sID,sName,GPA,sizeHS,DoB from student natural join CSaccept where cName = 'Berkeley' and major = 'CS' and sizeHS> 500;

- Select * from CSberkeley where GPA > 3.8;

- Drop view CSberkeley;

- Select sID, sName from CSberkeley;

- Update WeakStudent set GPA = GPA + 0.8 where sizeHS> 1000;

- Create or replace view AppCount as select sID, count(major) as 'No_of_App' from student group by sID;

- Update AppCount set No_of_App = 4 where sID = 234;

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